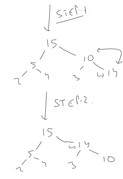

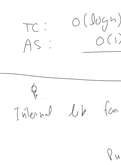

STOP as 14 is smaller than its parent

TC: O(logn)

AS: O(1)

Q

Internal lib for heaps → Priority Queue

Push → O(logn)

POP → O(logn)

(get minimum) top → O(1)

size → O(1)

empty → O(1)

Max-Heap Internal API

Insert n Vals in a max-Heap: O(nlogn)

(K-1) deletion: O(klogn)

Overall → O(nlogn)

AS → O(n)

Sorting Based API

TC → O(nlogn)

AS → O(1)

Min-Heap Internal API

K^th largest element = min of the K-largest elements

[3, 5, 4, 2, 9] K: 3

K-largest elements: 5, 4, 9

k k k

[3, 5, 4, 2, 9], K: 2

TC: O(nlog(k))

AS: O(K)